

Experiment No. 7

Experiment Name

Motion Estimation using Optical Flow Algorithms in Video Sequences

Aim

To implement optical flow algorithms for estimating the motion of objects between consecutive video frames and analyze motion patterns for dynamic scene understanding using Python and OpenCV.

Purpose

Optical Flow is a fundamental computer vision technique used to estimate the apparent motion of objects, surfaces, or camera movement by analyzing consecutive frames in a video sequence. Motion estimation plays a vital role in applications such as autonomous driving, video surveillance, robotics, gesture recognition, sports analytics, and augmented reality.

In this experiment, students will implement Lucas-Kanade Sparse Optical Flow and Farneback Dense Optical Flow algorithms to estimate object motion, visualize motion vectors, and compare the performance of both techniques on real-world video sequences.

Prerequisites

- Basic knowledge of Digital Image Processing
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), NumPy, and Matplotlib
- Understanding of video processing and image gradients
- Jupyter Notebook, Google Colab, or Visual Studio Code

Steps

1. Import the required Python libraries and load a video sequence or access a live webcam stream using OpenCV.
2. Read consecutive video frames and convert them to grayscale for optical flow computation.
3. Implement the Lucas-Kanade Optical Flow algorithm to estimate the motion of selected feature points across consecutive frames.
4. Visualize the motion vectors by drawing arrows or trajectories representing the movement of tracked points.

5. Implement the Farneback Dense Optical Flow algorithm to estimate motion for every pixel in the image.
6. Display the dense optical flow output using color-coded motion visualization to represent direction and magnitude.
7. Compare the performance of Lucas-Kanade and Farneback algorithms in terms of accuracy, computational complexity, and robustness.
8. Analyze the effect of object speed, lighting conditions, and camera motion on optical flow estimation.
9. Evaluate the suitability of each algorithm for different real-world applications such as object tracking, surveillance, and autonomous navigation.
10. Summarize the observations and discuss the significance of optical flow in dynamic scene analysis and motion-based computer vision systems.

Questions

1. What is Optical Flow? How is it used in computer vision?
2. Explain the working principle of the Lucas-Kanade Optical Flow algorithm.
3. What is Dense Optical Flow? How does it differ from Sparse Optical Flow?
4. Compare the Lucas-Kanade and Farneback optical flow algorithms.
5. What assumptions are made while computing optical flow?
6. What factors can affect the accuracy of optical flow estimation?
7. Mention five real-world applications of optical flow in computer vision.
8. Why are grayscale images generally used for optical flow computation?
9. What are the limitations of optical flow algorithms in real-world environments?
10. How does optical flow contribute to applications such as autonomous driving, video surveillance, and action recognition?